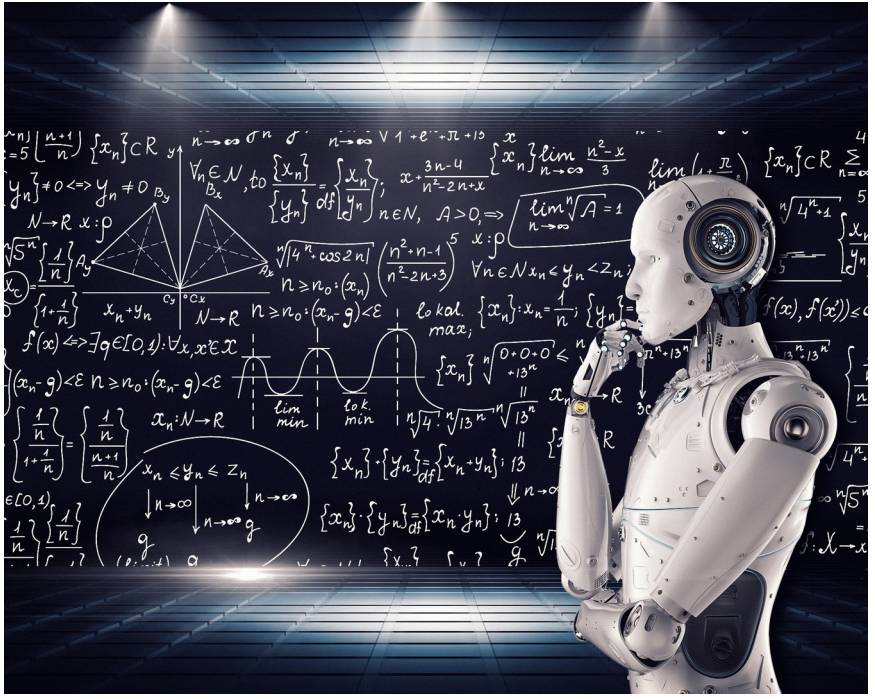


Decision Trees

Lecture 22
PHYS 4730 F26

Last Time

- Intro to M.L.
- Grouping or *Classification* Problems
- M.L. Jargon
- Quantifying impurity
- Lab: 1D Decision tree



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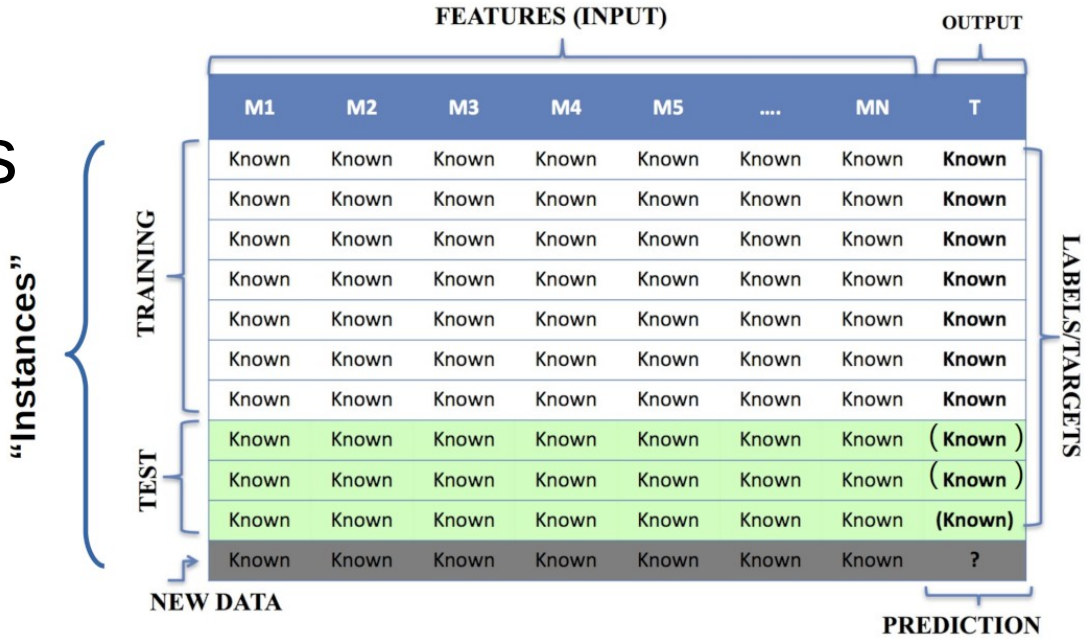
What Can We Do With It?



- **Recognize** an instance of a certain type. *e.g.* identify an animal as a cat.
- **Predict** some property or information on the basis of other information. *e.g.* predict whether my cat will like some food on the basis of other food she likes.
- **Group** similar objects. *e.g.* separate long-haired cats from short-haired cats in a collection of cats.
- **Simplify** information to it's essential nature. *e.g.* make the simplest drawing of a cat that is still recognizable as a cat.

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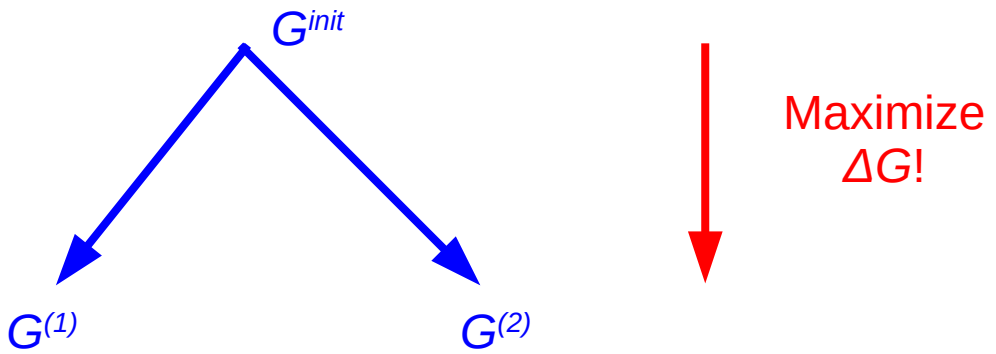


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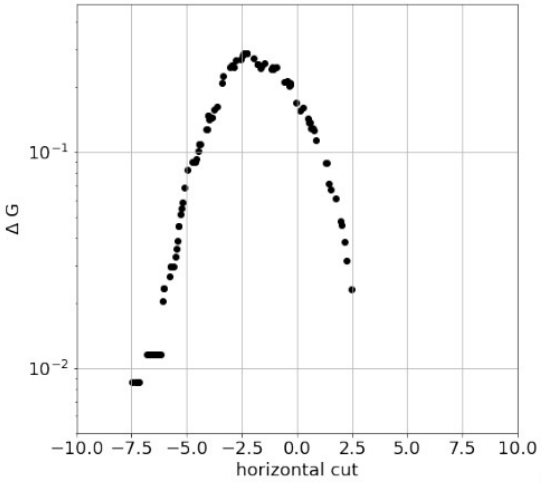
$$G = 1 - \sum_{i=1}^{N_{class}} f(i)^2$$

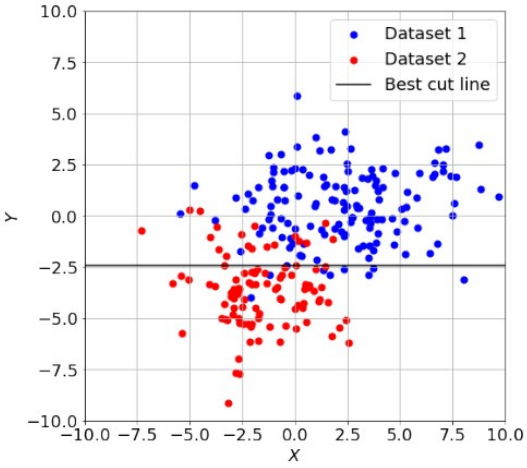
$$\Delta G = G^{init} - \left(G^{(1)} \frac{N^{(1)}}{N_{tot}} + G^{(2)} \frac{N^{(2)}}{N_{tot}} \right)$$



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Today's Lab: Implementation of Decision Trees in **scikit-learn**

	P_NAME	S_MASS	P_PERIOD	P_DISTANCE	P_HABITABLE
0	Kepler-736 b	0.86	3.601472	0.043718	0
1	Kepler-636 b	0.85	16.080661	0.118082	0
2	Kepler-887 c	1.19	7.638460	0.080420	0
3	Kepler-442 b	0.61	112.305300	0.409327	1
4	Kepler-772 b	0.98	12.992073	0.107408	0
5	Teegarden's Star b	0.09	4.910000	0.025200	1
6	K2-116 b	0.69	4.655411	0.048086	0
7	GJ 1061 c	0.12	6.689000	0.035000	1
8	HD 68402 b	1.12	1103.000000	2.180981	0
9	Kepler-1544 b	0.81	168.811170	0.557107	1
10	Kepler-296 e	0.50	34.142110	0.178202	1
11	Kepler-705 b	0.53	56.056054	0.231926	1
12	Kepler-445 c	0.18	4.871229	0.031746	0
13	HD 104067 b	0.62	55.806000	0.260000	0
14	GJ 4276 b	0.41	13.352000	0.087613	0
15	Kepler-296 f	0.50	63.336270	0.268885	1
16	Kepler-63 b	0.98	9.434150	0.088100	0
17	GJ 3293 d	0.42	48.134500	0.195336	1

Subset of Habitable Worlds Catalog

